
Does conditioning leave a structural memory?

From copper evidence to a stainless-steel test

Yinon Ashkenazy and collaborators

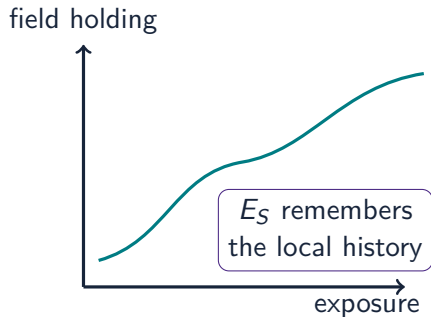
HUJI · CERN · Consorzio RFX

MeVArc 2026



Conditioning works—but what is being conditioned?

- ▶ Repeated high-field exposure raises voltage-holding capability.
- ▶ Conditioning follows **pulse count**, not simply breakdown count.
- ▶ A local state variable, E_S , captures the evolving field tolerance.



The missing link is a measurable material state that survives between pulses.

Source: Wuensch et al., Rep. Prog. Phys. (2026); Bjelland et al. (2026).

Dislocations are a plausible carrier of that memory

- ▶ Breakdown statistics and material trends track barriers to dislocation motion.
- ▶ The MDDF model treats initiation as a collective dislocation instability.
- ▶ Local TEM finds a ~ 200 nm dislocation-denuded layer in conditioned hard Cu.

Maxwell stress

$$\sigma_M = \frac{\epsilon_0 E^2}{2}$$

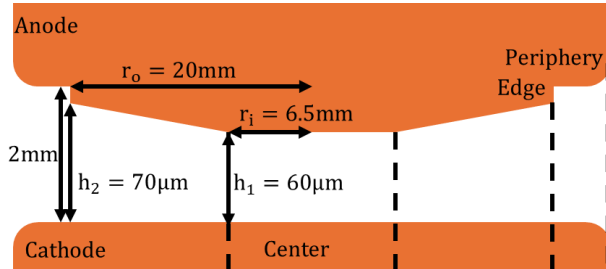
at 80 MV/m: **0.028 MPa**

tiny per pulse, repeated $\sim 10^9$ times

The hypothesis is cumulative, field-driven plastic rearrangement below macroscopic yield.

Source: Engelberg et al., PRL 2018 / PRAB 2019; Jacewicz et al., PRAB 2025.

One cathode turns field exposure into a controlled coordinate

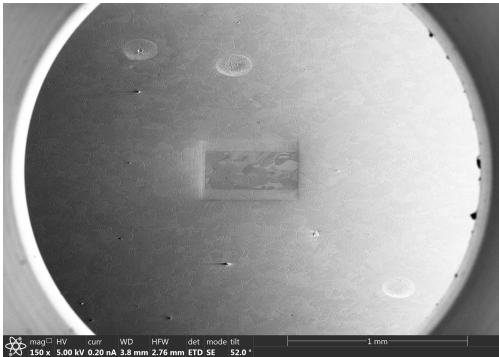


- ▶ OFE Cu, pulsed DC
- ▶ $1\mu\text{s}$ pulses at 1kHz
- ▶ Peak field $\sim 80\text{MV/m}$
- ▶ Nine EBSD regions: center, edge, periphery, and external reference

The spatial gradient supplies internal controls on the same specimen.

Source: Ashkenazy et al., submitted to PRAB (2026); geometry from Bjelland et al.

EBSD maps plastic activity far from breakdown craters



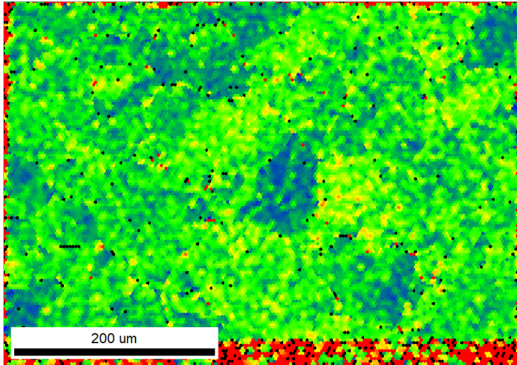
- ▶ 500 μm -wide maps; 3 μm step
- ▶ $\sim 24,000$ indexed points per ROI
- ▶ LAM, KAM, and LOS probe intragrain orientation gradients
- ▶ Misorientation is a proxy for geometrically necessary dislocation density

Cleaned ROI is hundreds of microns from visible craters.

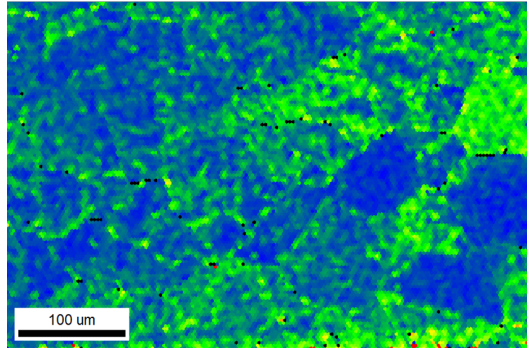
This separates cumulative field exposure from the local aftermath of an arc.

Source: cond26 manuscript, experimental methods.

High-field copper contains more intragrain curvature



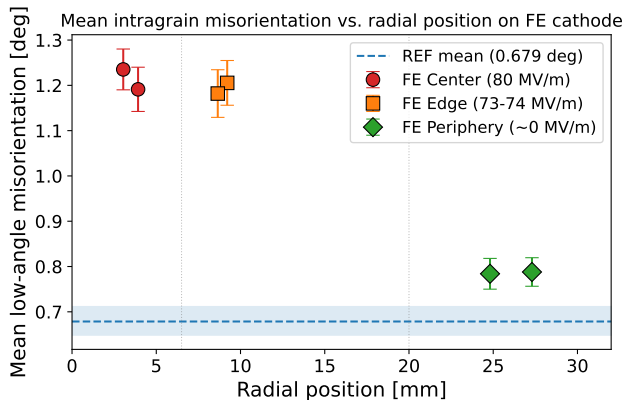
Field-exposed center



Unexposed reference

Source: Ashkenazy et al., submitted to PRAB (2026). Same LAM color scale.

The signal separates into three exposure tiers

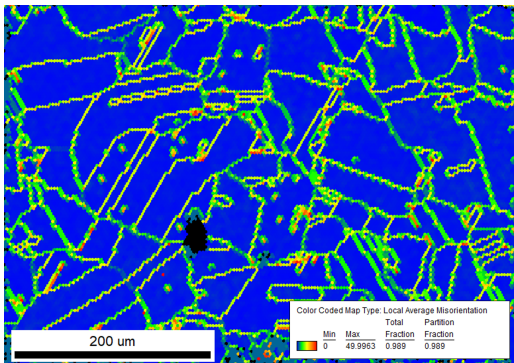


- ▶ Center and edge: $\sim 1.2^\circ$
- ▶ External reference: $\sim 0.68^\circ$
- ▶ High field is $\sim 75\%$ higher
- ▶ Grain size does not follow this hierarchy

The ordering follows field history, not merely sample position or grain size.

Source: Ashkenazy et al., submitted to PRAB (2026).

Field exposure populates a high-misorientation tail



LAM tail above 2°

~ 0.14 field-exposed

~ 0.016 reference

$\sim 8\times$ **increase**

All center/edge distributions

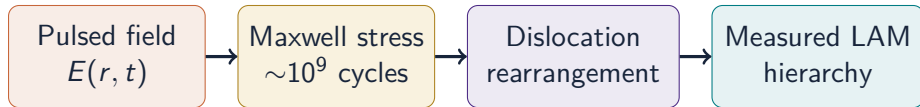
differ from unexposed references:

$D_{KS} > 0.25$.

Field exposure redistributes orientation; it is not a uniform offset.

Source: cond26 manuscript, distribution analysis.

A material state now shadows the conditioning state E_S



E_S : phenomenological memory



Misorientation: structural memory

The evolving subsurface dislocation population is a candidate physical basis for E_S .

Copper establishes the effect; it does not establish universality

What is established

- ▶ Large-area structural difference
- ▶ Three independent metrics
- ▶ Internal and external controls
- ▶ Spatial ordering consistent with E_S

What remains open

- ▶ Full depth profile
- ▶ Grain-orientation dependence
- ▶ More conditioned cathodes
- ▶ Cross-material generality

A stronger test changes both the material and the loading history.

Stainless steel is a deliberately difficult test

Copper

FCC, higher SFE
easier cross-slip
pulsed DC loading
 $\sim 10^9$ cycles

$\neq$

AISI 304 SS

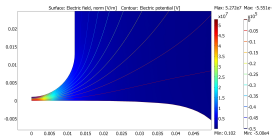
metastable austenite, lower SFE
planar slip / possible twinning
continuous DC exposure
possible $\gamma \rightarrow \varepsilon \rightarrow \alpha'$ transformation

A correlated SS response would support a mechanism broader than one alloy or pulse protocol.

The RFX electrodes provide a second field gradient



HVPTF sphere-plane assembly



Simulated field map

- ▶ Ø24 mm hemispherical cathode
- ▶ 1 mm gap
- ▶ two exposed electrodes
- ▶ ≥ 100 h at 60–80 MV/m
- ▶ two batch-matched references

Apex-to-periphery variation enables an internal field-correlation test.

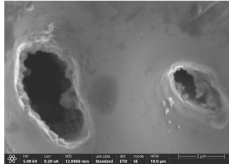
Source: Consorzio RFX electrode drawing and electrostatic simulation; RFX project status, July 2026.

The first SEM survey reveals a new surface signature

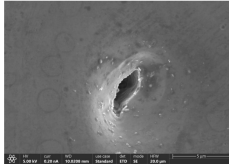
Sample 1-25 ref



Sample 3-25 tested



Sample 4-25 tested



- ▶ Reference: cleaner, rounded cavities
- ▶ Tested: irregular erupted sites
- ▶ Typical scale: 1–7 µm
- ▶ No clear molten rim or recast layer

The morphology is consistent with ruptured near-surface inclusions—not yet proof of a field mechanism.

Source: HUJI SEM survey of RFX samples #1-25, #3-25, and #4-25 (May 2026).

Early EBSD behavior may already be diagnostic

Reference #1-25

- ▶ good diffraction patterns
- ▶ LAM and grain statistics extracted
- ▶ Cu analysis pipeline transfers

Field-exposed #4-25

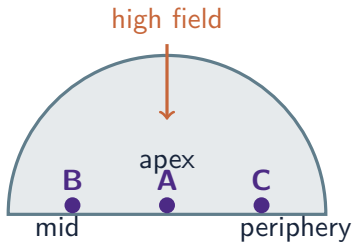
- ▶ poor pattern quality
- ▶ higher current / longer exposure did not recover it
- ▶ cause is unresolved

Three hypotheses: preparation variation conditioning-induced defects
oxide/contamination overlayer

Pattern degradation is evidence only if it repeats and follows the field gradient.

Source: RFX project status and HUJI EBSD update, 13 July 2026.

The decisive comparison is within each exposed hemisphere



Conditioning signature

Pattern quality and/or misorientation vary monotonically with local field on both exposed samples.

Preparation signature

Differences are sample-wide, non-monotonic, or also appear on matched references.

The gradient converts an ambiguous sample contrast into a falsifiable spatial prediction.

Depth profiling can manufacture the signal we seek

Ga damage A 30 kV FIB

can damage a layer comparable to the EBSD information depth.

Redeposition Serial

milling can mix material and obscure the true depth coordinate.

False martensite Ion

irradiation can transform metastable austenite to BCC, mimicking α' formation.

Pilot-first control: identical preparation on exposed and reference material; low-energy final polish; dose bookkeeping; go/no-go before the full $\sim 1\ \mu\text{m}$ campaign.

Preparation artifacts are part of the experiment, not a post-processing detail.

Source: RFX artifact review; Babu et al., Acta Mater. 120, 391 (2016).

Every outcome constrains the conditioning mechanism

SS tracks field

Supports a cross-material structural state despite different slip physics and loading.

SS changes differently

Separates material-specific carriers: planar slip, twins, phase change, or inclusions.

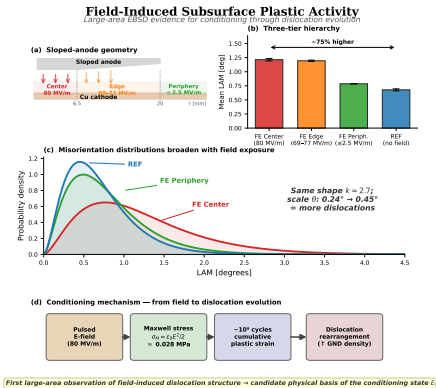
No resolved change

Bounds the accessible depth, dose, and sensitivity of dislocation-based models.

The next result is not simply “positive or negative”; it is a mechanism discriminator.

Conditioning appears to leave a subsurface memory

1. In copper, large-area misorientation is $\sim 75\%$ higher in high-field regions.
2. Its spatial hierarchy follows the conditioning-state profile E_S .
3. Stainless steel tests whether that memory survives a change of alloy and loading mode.



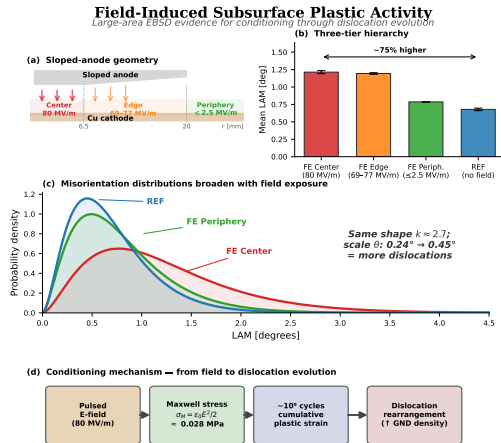
We can now ask what conditioning stores—and where it stores it.

Backup: RFX sample and data status

Sample	Role	Surface	Current status
#1-25	reference	electropolished	SEM + good EBSD
#2-25	reference	electropolished	follow-up acquisition planned
#3-25	field-exposed	≥ 100 h HV	SEM; EBSD follow-up
#4-25	field-exposed	≥ 100 h HV	SEM; poor initial EBSD

- ▶ Outstanding: full conditioning report; numerical $E(\theta)$ vector; alloy certificate; electropolishing protocol.
- ▶ Needed for interpretation: quantitative EBSD quality metrics and raw patterns; matched EDS overlay checks.

Backup: copper result in one figure



First large-area observation of field-induced dislocation structure → candidate physical basis of the conditioning state E_S